Rabi Oscillations: Theory and Numerical Simulation

Supporting document for `Rabi_Oscillations.ipynb`

1 Overview

This document explains the physics and numerics behind the notebook `Rabi_Oscillations.ipynb`, which simulates a two-level quantum system driven by an oscillating field. It covers four things: (i) Rabi oscillations, (ii) time-dependent perturbation theory, (iii) the Euler method for integrating the Schrödinger equation, and (iv) a section-by-section walkthrough of the code. Throughout, units are chosen so that $\hbar = 1$, exactly as in the code, except in Section 3.5, where $\hbar$ is kept explicit.

2 Rabi oscillations

2.1 Physical picture

A Rabi oscillation is the coherent, periodic exchange of population between two quantum states when a system is driven by an oscillating field whose frequency is close to the transition frequency. A two-level atom in a laser field, a spin in an RF field (NMR/ESR), and a superconducting qubit driven by a microwave pulse are all examples. Instead of settling into a steady mixture, the system keeps cycling: state $|0\rangle \rightarrow$ superposition $\rightarrow$ state $|1\rangle \rightarrow$ superposition $\rightarrow$ back to $|0\rangle$.

2.2 The model used in the notebook

The Hamiltonian is $H(t) = H_0 + V(t)$, where

$$H_0 = \delta\, \sigma_z, \tag{1}$$

$$V(t) = \mu\, e^{i\omega t} + \mu^{\dagger} e^{-i\omega t}, \tag{2}$$

$$\mu = \frac{\Omega}{2}\, \sigma_x. \tag{3}$$

Because μ is Hermitian here (real and symmetric), the perturbation simplifies to

$$V(t) = \Omega \cos(\omega t)\, \sigma_x. \tag{4}$$

The basis states are $|0\rangle = (1, 0)^T$ with energy $+\delta$ and $|1\rangle = (0, 1)^T$ with energy $-\delta$. The energy gap, and therefore the resonant (transition) frequency, is

$$\omega_0 = 2\delta. \tag{5}$$

Note that δ is half the splitting, not the whole splitting. The drive σ_x has zero diagonal elements and couples $|0\rangle$ to $|1\rangle$, which is what drives transitions.

2.3 Derivation with the rotating-wave approximation (RWA)

Write the state as $\psi(t) = c_0(t)\, e^{-i\delta t}|0\rangle + c_1(t)\, e^{+i\delta t}|1\rangle$, which removes the fast evolution due to H_0. Substituting into $i\dot{\psi} = H\psi$ gives

$$i\dot{c}_0 = \Omega\cos(\omega t)\, e^{+i\omega_0 t}\, c_1, \tag{6}$$

$$i\dot{c}_1 = \Omega\cos(\omega t)\, e^{-i\omega_0 t}\, c_0. \tag{7}$$

Expanding $\cos(\omega t) = (e^{i\omega t} + e^{-i\omega t})/2$, the equation for c_1 contains two terms:

$$i\dot{c}_1 = \frac{\Omega}{2}\Big[e^{i(\omega-\omega_0)t} + e^{-i(\omega+\omega_0)t}\Big]c_0. \tag{8}$$

The first term oscillates slowly (at the detuning $\Delta = \omega - \omega_0$). The second, the counter-rotating term, oscillates at $\omega + \omega_0$, which is very fast and averages to nearly zero. The RWA drops it, leaving

$$i\dot{c}_1 = \frac{\Omega}{2}\, e^{i\Delta t}\, c_0, \tag{9}$$

$$i\dot{c}_0 = \frac{\Omega}{2}\, e^{-i\Delta t}\, c_1. \tag{10}$$

Starting from $c_0(0) = 1$, $c_1(0) = 0$, the probability of finding the system in $|1\rangle$ is the *Rabi formula*:

$$P_1(t) = \frac{\Omega^2}{\Omega^2 + \Delta^2}\, \sin^2\left(\frac{\sqrt{\Omega^2 + \Delta^2}\, t}{2}\right). \tag{11}$$

2.4 Key results

- **On resonance** ($\Delta = 0$): $P_1(t) = \sin^2(\Omega t/2)$. The population swings fully between 0 and 1. Here Ω is the Rabi frequency (in the convention where the drive is written $\Omega\cos(\omega t)\,\sigma_x$).
- **Off resonance:** the oscillation is faster, $\sqrt{\Omega^2 + \Delta^2}$, but its amplitude shrinks to $\Omega^2/(\Omega^2 + \Delta^2)$, so transfer is incomplete.
- **π-pulse:** driving for $t = \pi/\Omega$ moves the system completely from $|0\rangle$ to $|1\rangle$.
- **2π-pulse:** driving for $t = 2\pi/\Omega$ returns it to $|0\rangle$ (with a global phase of -1 on the amplitude).
- **Period of the population oscillation:** $2\pi/\Omega$.

Conventions differ between textbooks (some write the drive as $(\Omega/2)\cdot 2\cos(\omega t)\,\sigma_x$, some define Ω as the full splitting of the dressed states), so always check how Ω enters the Hamiltonian before comparing formulas.

3 Time-dependent perturbation theory

3.1 Setting

When the Hamiltonian splits as $H = H_0 + V(t)$, with H_0 exactly solvable (eigenstates $|n\rangle$, energies E_n) and $V(t)$ small, we expand the evolution in powers of V. Expand the state in the unperturbed basis, $|\psi(t)\rangle = \sum_n c_n(t)\, e^{-iE_n t}|n\rangle$. The Schrödinger equation becomes an exact set of coupled equations:

$$i\dot{c}_f(t) = \sum_n V_{fn}(t)\, e^{i\omega_{fn} t}\, c_n(t), \qquad \omega_{fn} = E_f - E_n, \quad V_{fn} = \langle f|V|n\rangle. \tag{12}$$

3.2 The Dyson series

In the interaction picture the evolution operator satisfies $i\,dU_I/dt = V_I(t)\,U_I$, which integrates to the Dyson series:

$$U_I(t) = 1 - i\int_0^t V_I(t_1)\,dt_1 - \int_0^t dt_1 \int_0^{t_1} dt_2\, V_I(t_1)V_I(t_2) + \cdots \tag{13}$$

If the system starts in $|i\rangle$, the amplitude to be found in $|f\rangle$ is $c_f = c_f^{(0)} + c_f^{(1)} + c_f^{(2)} + \cdots$, and to **first order** ($f \neq i$):

$$c_f^{(1)}(t) = -i\int_0^t \langle f|V(t')|i\rangle\, e^{i\omega_{fi}t'}\,dt', \tag{14}$$

$$P_{i\to f}(t) \approx \left|c_f^{(1)}(t)\right|^2. \tag{15}$$

This is valid only while the transition probability stays small, $|c_f| \ll 1$.

3.3 Harmonic perturbation

For $V(t) = \mu\, e^{i\omega t} + \mu^\dagger e^{-i\omega t}$, the first-order integral can be done in closed form. Keeping the term that can be resonant (absorption or emission, depending on the sign of ω_{fi}), the probability is

$$P_{i\to f}(t) \approx |\mu_{fi}|^2\, \frac{\sin^2\big((\omega_{fi} \pm \omega)\,t/2\big)}{\big((\omega_{fi} \pm \omega)/2\big)^2}. \tag{16}$$

This is sharply peaked when the drive frequency matches the transition frequency. For a transition into a continuum of final states, the $\sin^2$ factor becomes a delta function at long times and the probability grows linearly in time, giving **Fermi's golden rule**:

$$\Gamma = 2\pi\,|\langle f|V|i\rangle|^2\,\rho(E_f). \tag{17}$$

3.4 How it connects to Rabi oscillations

For the notebook, $\langle 1|\mu|0\rangle = \Omega/2$ and the drive is resonant, so the first-order amplitude is

$$c_1^{(1)}(t) \approx -i\,\frac{\Omega}{2}\,t \quad\Longrightarrow\quad P_1 \approx \left(\frac{\Omega t}{2}\right)^2. \tag{18}$$

This agrees with the short-time expansion of the exact Rabi result, $\sin^2(\Omega t/2) \approx (\Omega t/2)^2$. However, first-order theory predicts unbounded quadratic growth, which is impossible for a probability. The Rabi oscillation is what you get when all orders of the Dyson series are resummed: the population saturates at 1 and turns around. So perturbation theory explains the beginning of the transition, and the Rabi formula (or direct numerical integration, as in the notebook) describes the full dynamics.

3.5 Derivation of the transition probability (interaction picture)

This section derives $P_{if}(t)$ step by step from the Schrödinger equation in the interaction picture, following the lecture-board derivation. Here $\hbar$ is kept explicit; setting $\hbar = 1$ recovers the formulas of the previous subsections.

Step 0: Definitions. With $H = H_0 + V(t)$, the interaction-picture state and perturbation are

$$|\psi(t)\rangle_I = e^{iH_0t/\hbar}\,|\psi(t)\rangle_S, \qquad \hat{V}_I(t) = e^{iH_0t/\hbar}\,\hat{V}(t)\,e^{-iH_0t/\hbar}. \tag{19}$$

Step 1: Equation of motion. In this picture the state evolves under the perturbation alone:

$$i\hbar \frac{d}{dt}|\psi(t)\rangle_I = \hat{V}_I(t)\,|\psi(t)\rangle_I. \tag{20}$$

Step 2: Evolution operator. Write the state at time t in terms of the state at the initial time t_i:

$$|\psi(t)\rangle_I = U_I(t,t_i)\,|\psi(t_i)\rangle_I. \tag{21}$$

Substituting into Step 1 gives

$$i\hbar \frac{d}{dt}\Big[U_I(t,t_i)\,|\psi(t_i)\rangle_I\Big] = \hat{V}_I(t)\,U_I(t,t_i)\,|\psi(t_i)\rangle_I. \tag{22}$$

Since $|\psi(t_i)\rangle$ is arbitrary, the same equation holds for the operator itself:

$$i\hbar \frac{d}{dt}U_I(t,t_i) = \hat{V}_I(t)\,U_I(t,t_i), \qquad U_I(t_i,t_i) = 1. \tag{23}$$

Step 3: Integral equation. Integrating from t_i to t gives an exact but implicit equation, because U_I appears on both sides:

$$U_I(t,t_i) = 1 - \frac{i}{\hbar}\int_{t_i}^{t} \hat{V}_I(t')\,U_I(t',t_i)\,dt'. \tag{24}$$

This form can be solved numerically step by step (the Euler method of Section 4 is the simplest way), or iteratively, as in Step 4.

Step 4: Iteration gives the Dyson series. Substituting the right-hand side back into itself again and again produces a series in powers of $\hat{V}_I$, with the times nested so that $t_i \le t'' \le t' \le t$:

$$\begin{aligned} U_I(t,t_i) = 1 &- \frac{i}{\hbar}\int_{t_i}^{t} \hat{V}_I(t')\,dt' + \left(\frac{-i}{\hbar}\right)^2 \int_{t_i}^{t} \hat{V}_I(t')\,dt' \int_{t_i}^{t'} \hat{V}_I(t'')\,dt'' \\ &+ \cdots + \left(\frac{-i}{\hbar}\right)^n \int_{t_i}^{t} dt^{(1)} \cdots \int_{t_i}^{t^{(n-1)}} dt^{(n)}\, \hat{V}_I(t^{(1)}) \cdots \hat{V}_I(t^{(n)}) + \cdots \end{aligned} \tag{25}$$

Step 5: Transition amplitude to second order. For a transition from $|\psi_i\rangle$ to a different state $|\psi_f\rangle$ the identity term drops out, because $\langle\psi_f|\psi_i\rangle = 0$. Keeping terms up to second degree, with $|n\rangle$ running over intermediate states:

$$\begin{aligned} \langle\psi_f|U_I(t,t_i)|\psi_i\rangle \approx\; &- \frac{i}{\hbar}\int_{t_i}^{t} \langle\psi_f|\hat{V}_I(t')|\psi_i\rangle\,dt' \\ &+ \left(\frac{-i}{\hbar}\right)^2 \sum_n \int_{t_i}^{t} dt' \int_{t_i}^{t'} dt''\, \langle\psi_f|\hat{V}_I(t')|n\rangle\langle n|\hat{V}_I(t'')|\psi_i\rangle. \end{aligned} \tag{26}$$

The transition probability is the squared modulus of this amplitude:

$$P_{if}(t) = \Big|\langle\psi_f|U_I(t,t_i)|\psi_i\rangle\Big|^2. \tag{27}$$

Step 6: First-order result. The interaction-picture matrix element only adds a phase, with $\omega_{fi} = (E_f - E_i)/\hbar$:

$$\langle\psi_f|\hat{V}_I(t)|\psi_i\rangle = \langle\psi_f|\hat{V}(t)|\psi_i\rangle\, e^{i\omega_{fi}t}. \tag{28}$$

Taking $t_i = 0$ and keeping only the first-order term gives the boxed result from the board:

$$\boxed{P_{if}(t) \approx \left| -\frac{i}{\hbar}\int_0^t \langle\psi_f|\hat{V}(t')|\psi_i\rangle\, e^{i\omega_{fi}t'}\, dt' \right|^2} \tag{29}$$

The square applies to the whole integral, which is the amplitude. The board shows the squared bars around the matrix element inside the integral, which looks like a notation slip, so it is written here as the modulus squared of the full amplitude.

For the harmonic perturbation $V(t) = \mu\, e^{i\omega t} + \mu^\dagger e^{-i\omega t}$ used in the notebook, this integral gives the resonance formulas of the harmonic-perturbation subsection and, on resonance, the short-time result $P_1 \approx (\Omega t/2)^2$. Note that the coefficients c_0 and c_1 in the RWA derivation are the interaction-picture amplitudes of this derivation, while the notebook itself integrates the Schrödinger-picture state directly.

4 The Euler method

4.1 The method

For an ordinary differential equation $y' = f(t, y)$, the (forward) Euler method advances the solution by one step of size h using the slope at the start of the step:

$$y_{n+1} = y_n + h\, f(t_n, y_n). \tag{30}$$

It is the first-order Taylor expansion of the solution. The local error per step is $O(h^2)$ and the accumulated (global) error is $O(h)$.

4.2 Applied to the Schrödinger equation

The Schrödinger equation is $i\dot{\psi} = H(t)\psi$, i.e. $\dot{\psi} = -iH(t)\psi$. Euler's method gives

$$\psi_{n+1} = \psi_n + dt\,(-i)\,H(t_n)\,\psi_n = \big(I - iH(t_n)\,dt\big)\,\psi_n. \tag{31}$$

This is exactly the update line in the notebook.

4.3 The unitarity problem

The exact evolution operator is unitary, so the norm of ψ is conserved. The Euler propagator is not:

$$\big(I - iH\,dt\big)^\dagger\big(I - iH\,dt\big) = I + H^2\,dt^2. \tag{32}$$

So for an eigenvalue E of H, the norm grows by a factor $\sqrt{1 + E^2\,dt^2}$ every step. In this notebook, the norm of the raw, un-normalised Euler solution grows to roughly 25 by $t = 4\pi$ (checked by running the loop without the normalisation line). That is why the code divides by `np.linalg.norm` after each step: it restores the total probability to 1.

Renormalisation fixes the norm but not the phase error, so it is a patch rather than a cure. The method is still adequate here because the step size is small compared with the fastest time scales.

4.4 Step size in this notebook

- $dt = T/(N-1) = 4\pi/999 \approx 0.0126$
- the drive period is $2\pi/\omega = 0.5$, so there are about 40 steps per drive cycle
- the phase advanced per step by the drive is $\omega\, dt \approx 0.16$ rad

4.5 Better alternatives

- **Matrix-exponential step:** $\psi_{n+1} = \exp\big(-iH(t_n + dt/2)\, dt\big)\, \psi_n$ is exactly unitary (use `scipy.linalg.expm`).
- **Crank–Nicolson (Cayley) scheme:** $(I + iH\, dt/2)\, \psi_{n+1} = (I - iH\, dt/2)\, \psi_n$, unitary and second-order.
- **Runge–Kutta 4 (RK4):** much higher accuracy per step but still not exactly unitary.
- **Magnus expansion:** a family of unitary integrators designed for time-dependent Hamiltonians.

5 Code walkthrough

The full listing is given in the appendix. Each part is explained here.

5.1 Imports

```
import numpy as np
import matplotlib.pyplot as plt
```

NumPy handles complex arrays and matrix products; Matplotlib draws the plot.

5.2 The perturbation function

```
def V_t(mu, t, omega=1):
    return mu * np.exp(1j*omega*t) + np.conj(mu.T) * np.exp(-1j*omega*t)
```

This implements $V(t) = \mu\, e^{i\omega t} + \mu^\dagger e^{-i\omega t}$. The Hermitian conjugate $\mu^\dagger$ is written as `np.conj(mu.T)`, so $V(t)$ is Hermitian for every t, as a Hamiltonian must be.

5.3 Operators and initial state

```
sigma_x = np.array([[0,1],[1,0]], dtype=complex)
sigma_z = np.array([[1,0],[0,-1]], dtype=complex)
psi0 = np.array([1,0], dtype=complex)
ket1 = np.array([0,1], dtype=complex)
```

σ_x and σ_z are the Pauli matrices. `psi0` $= |0\rangle$ is the initial state, with the system fully in the first level. `ket1` $= |1\rangle$ is the target state whose population is plotted.

5.4 Time grid and parameters

```
N = 1000
T = 4*np.pi
t = np.linspace(0, T, N)
delta = 2*np.pi # half of the level splitting
```

```
omega = 2*delta # drive frequency = 4*pi
Omega = 0.5*np.pi # Rabi frequency = pi/2
```

- $N = 1000$ time points over a total time $T = 4\pi \approx 12.57$.
- $\delta = 2\pi$, so the level splitting is $2\delta = 4\pi$, and $\omega = 2\delta = 4\pi$ equals it. The drive is therefore **exactly on resonance** ($\Delta = 0$).
- $\Omega = \pi/2 \approx 1.57$ is much smaller than $\omega_0 = 4\pi$, so the RWA is valid and the Rabi formula applies.

5.5 Hamiltonian pieces

```
H0 = sigma_z*delta
mu = (Omega/2.0) * sigma_x
dt = t[1]-t[0]
```

$H_0 = \delta\sigma_z$ is the unperturbed Hamiltonian, $\mu = (\Omega/2)\sigma_x$ sets the coupling strength, and `dt` is the (uniform) time step.

5.6 The Euler time loop

```
psi = []
psi.append(psi0)
for time in t:
    H_t = H0 + V_t(mu, time, omega)
    psi_next = psi[-1] + dt*(-1j)*(H_t)@psi[-1]
    psi.append(psi_next/np.linalg.norm(psi_next))
```

At every time point the code (1) builds the full Hamiltonian $H(t) = H_0 + V(t)$, (2) takes an Euler step $\psi + dt \cdot (-i)H\psi$, and (3) renormalises the result and stores it. The list `psi` begins with the initial state, so it holds $N + 1$ states after the loop; entry k corresponds to time t_k.

5.7 Converting states to probabilities

```
for i in range(len(psi)):
    psi[i] = abs(ket1@psi[i])**2
```

The overlap $\langle 1|\psi\rangle$ is the amplitude on $|1\rangle$, and its squared modulus is the population P_1. Here the list `psi` is overwritten with real numbers. (`ket1` is real, so the plain `@` product equals the inner product; for a general complex bra use `np.vdot(ket1, psi[i])`.)

5.8 Plotting

```
plt.plot(t, psi[:N], color='red', label='|<1|psi(t)>|^2')
plt.ylim(0,1); plt.xlim(0,T)
plt.legend(); plt.grid(); plt.tight_layout(); plt.show()
```

The slice `psi[:N]` discards the one extra state computed beyond the final time so that the x and y arrays have equal length.

6 What the output shows and how it was checked

The plot is a red curve that oscillates smoothly between 0 and 1. For $\Omega = \pi/2$, the resonant Rabi formula predicts $P_1 = \sin^2(\pi t/4)$, with:

- a period of $2\pi/\Omega = 4$,
- full transfer (π-pulse) at $t = \pi/\Omega = 2$, and further maxima at $t = 6$ and $t = 10$,
- zeros at $t = 0, 4, 8, 12$, i.e. about $\pi \approx 3.14$ full cycles in $T = 4\pi$.

The notebook logic was compared with the analytic result and with a high-accuracy reference solution (a unitary midpoint matrix-exponential integration with 200,000 steps):

- Euler peak population ≈ 0.998 at $t \approx 2.0$, 6.0, 10.0 (reference ≈ 0.999).
- Euler vs. the RWA formula: RMS difference ≈ 0.02, maximum ≈ 0.06.
- The small remaining differences come from two sources: Euler discretisation error, and small fast ripples that are real physics (the counter-rotating terms, known as the Bloch–Siegert effect) which the RWA formula omits.

7 Possible extensions

- Overlay the analytic curve $\sin^2(\Omega t/2)$ on the plot to show agreement.
- Replace the Euler step by a matrix-exponential step to keep the evolution exactly unitary and remove the renormalisation.
- Sweep the drive frequency ω to show detuning: reduced amplitude and faster oscillation off resonance, and the Lorentzian-shaped resonance peak.
- Increase Ω towards ω_0 to see the breakdown of the RWA.
- Rename the reused variable `psi` (states, then probabilities) to avoid confusion, and use a bra such as `np.vdot` for general complex states.

A Full code

```
import numpy as np
import matplotlib.pyplot as plt

# Defining the time dependent perturbation
def V_t(mu, t, omega=1):
    return mu * np.exp(1j*omega*t) + np.conj(mu.T) * np.exp(-1j*omega*t)

sigma_x = np.array([[0,1],[1,0]], dtype=complex)
sigma_z = np.array([[1,0],[0,-1]], dtype=complex)

psi0 = np.array([1,0], dtype=complex)
ket1 = np.array([0,1], dtype=complex)
N = 1000
T = 4*np.pi
t = np.linspace(0,T,N)
psi = []
psi.append(psi0)

delta = 2 * np.pi # Energy splitting (H0 scale)
omega = 2*delta # Driving frequency
```

```
Omega = 0.5*np.pi

H0 = sigma_z*delta
mu = (Omega / 2.0) * sigma_x
dt = t[1]-t[0]
for time in t:
    H_t = H0 + V_t(mu, time, omega)
    psi_next = psi[-1] + dt*(-1j)*(H_t)@psi[-1]
    psi.append(psi_next/np.linalg.norm(psi_next))
for i in range(len(psi)):
    psi[i] = abs(ket1@psi[i])**2
plt.plot(t, psi[:N], color='red', label='|<1|psi(t)>|^2')
plt.ylim(0,1)
plt.xlim(0,T)
plt.legend()
plt.grid()
plt.tight_layout()
plt.show()
```